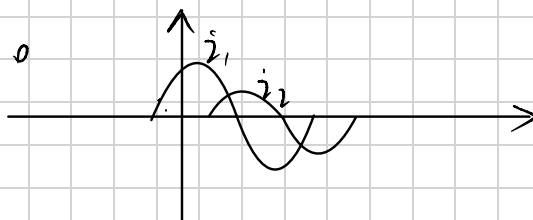


习题 3

$$\begin{aligned} 3-1. \quad i_{1m} &= 31.1 \text{ A} & i_{2m} &= 10.5 \text{ A} \\ i_1 &= 21.99 \text{ A} & i_2 &= 10 \text{ A} \\ f_1 &= 50 \text{ rad/s} & f_2 &= 50 \text{ rad/s} \\ T_1 &= 0.02 \text{ s} & T_2 &= 0.02 \text{ s} \\ \varphi_1 &= 20^\circ & \varphi_2 &= -70^\circ \end{aligned}$$

$$\Delta\varphi = 90^\circ$$



$$\begin{aligned} 3-3 \quad u_1 &= 220\sqrt{2} \sin(\omega t + 30^\circ) \text{ V} & u_2 &= 100\sqrt{2} \sin(\omega t) \text{ V} \\ I_1 &= 15\sqrt{2} \sin(\omega t + 90^\circ) \text{ A} & I_2 &= 30 \sin(\omega t - 45^\circ) \text{ A} \end{aligned}$$

$$3-5 \quad \dot{U}_1 = -50 \text{ j}$$

$$\begin{aligned} \cancel{R_2} - jX_{C2} &= \frac{\dot{U} - \dot{U}_1}{\dot{I}} = \frac{55 - 25\text{j}}{6 \angle 0^\circ} \\ &= \frac{6.042 \angle -24.44^\circ}{\angle 0^\circ} \\ &= 12.08 \angle -24.44^\circ \\ &= 11 - 5\text{j} \end{aligned}$$

$$\therefore R_2 = 11 \Omega \quad X_{C2} = 5$$

$$3-8 \quad \text{设 } \dot{U} = 220 \angle 0^\circ \text{ V}$$

$$\dot{I}_1 = \frac{220 \angle 0^\circ}{6 - 8\text{j}} = \frac{220 \angle 0^\circ}{10 \angle -53^\circ} = 22 \angle 53^\circ$$

$$\dot{I}_2 = \frac{220 \angle 0^\circ}{6 + 8\text{j}} = \frac{220 \angle 0^\circ}{10 \angle 53^\circ} = 22 \angle -53^\circ$$

$$\dot{I} = \dot{I}_1 + \dot{I}_2 = 13.2 + 17.6\text{j} + 13.2 - 17.6\text{j} = 26.4 \text{ A}$$

$$I = |\dot{I}| = 26.4 \text{ A}$$

$$\begin{aligned} \dot{U}_{ab} &= \dot{U}_a - \dot{U}_b \\ &= (\dot{U} - \dot{U}_{R1}) - (\dot{U} - \dot{U}_{R2}) \end{aligned}$$

$$\begin{aligned}
&= \dot{U}_{R_2} - \dot{U}_{R_1} \\
&= \dot{I}_2 R_2 - \dot{I}_1 R_1 \\
&= 6 (\dot{I}_2 - \dot{I}_1) \\
&= 6 \times (-35.2 j) \\
&= 211.2 \angle -90^\circ \text{ V}
\end{aligned}$$

$$U_{ab} = |\dot{U}_{ab}| = 211.2 \text{ V}$$

故 $U_{ab} = 211.2 \text{ V} \quad I = 26.4 \text{ A}$

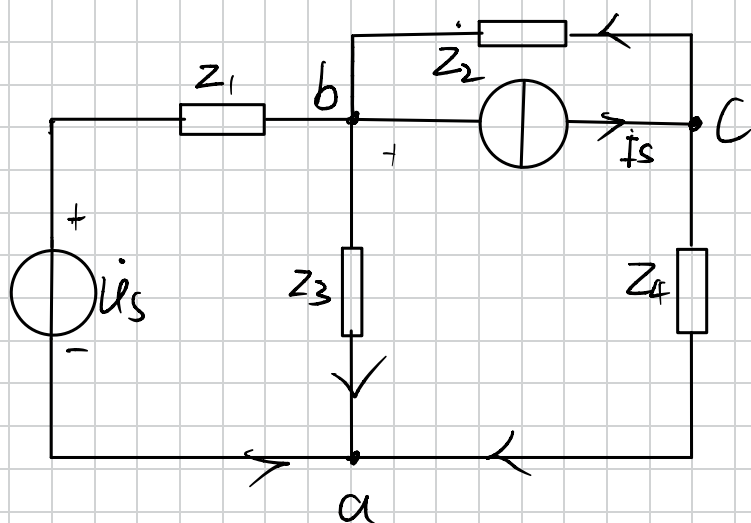
3-9.

$$\begin{aligned}
Z_{23} &= \frac{Z_2 Z_3}{Z_2 + Z_3} = \frac{(1 + 3j)(1 - 2j)}{1 + 3j + 1 - 2j} = \frac{5\sqrt{2} \angle 8.13^\circ}{\sqrt{5} \angle 26.57^\circ} \\
&= \sqrt{10} \angle -18.44^\circ \Omega
\end{aligned}$$

$$\begin{aligned}
\dot{U} &= \dot{I} (Z_1 + Z_{23}) = 1 \angle 0^\circ (5 + 3j + 3 - 1j) \\
&= 1 \angle 0^\circ \cdot 8.246 \angle 14.05^\circ \\
&= 8.246 \angle 14.05^\circ
\end{aligned}$$

3-10 ? ★

节点电位法:



如图节点 a, b, c , 设 a 节点电势为 0
 b, c 节点电势为 U_b, U_c

由 KCL 定律

$$\frac{U_c - U_b}{Z_2} - \dot{I}_s + \frac{U_c}{Z_4} = 0 \quad \text{--- ①}$$

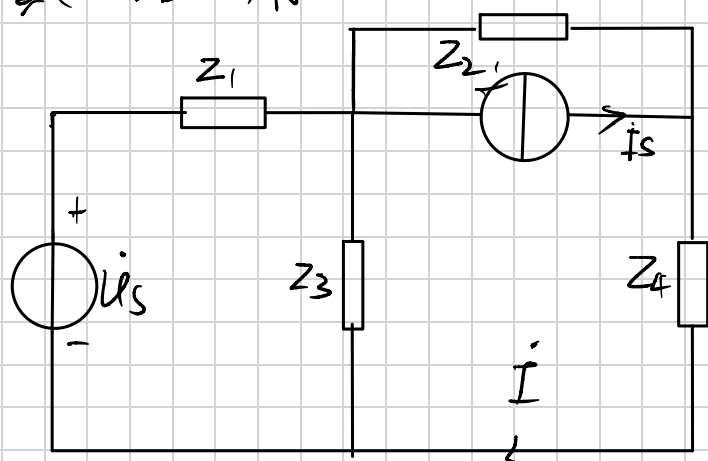
$$\frac{U_b - \dot{U}_s}{Z_1} + \frac{U_b}{Z_3} + \dot{I}_s - \frac{U_c - U_b}{Z_2} = 0 \quad \text{--- ②}$$

联立 ① ② 得

$$\dot{U}_c = 79.06 \angle 18.4^\circ \text{ V}$$

$$\dot{I} = \frac{\dot{U}_c}{Z_4} = 15.8 \angle -71.6^\circ \text{ A}$$

~~戴~~ 维南定理

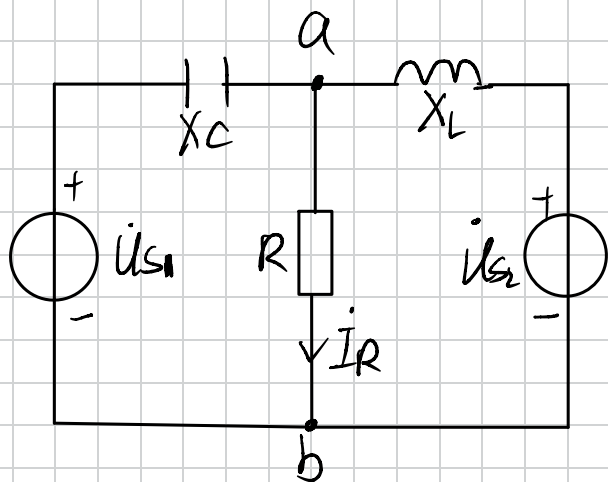


$$\begin{aligned}
 U_{DC} &= \dot{U}_S - \frac{Z_3}{Z_1 + Z_3} + \dot{I}_S Z_2 \\
 &= 50 \angle 0^\circ + 100\sqrt{2} \angle -135^\circ \\
 &= 50 - 100 - 100j \\
 &= -50 - 100j
 \end{aligned}$$

$$Z = \frac{Z_1 Z_3}{Z_1 + Z_3} + Z_2 = \frac{-100}{20} - 10j = 5 - 10j$$

$$\dot{I} = \frac{U_{DC}}{Z_4 + Z} = \frac{U_{DC}}{5 - 10j} = 15.8 \angle 71.6^\circ \text{ A}$$

3-11.



$$\dot{U}_{S1} = 100 \angle 0^\circ \text{ V}$$

$$\dot{U}_{S2} = 100 \angle 45^\circ \text{ V}$$

$$R = 5 \Omega$$

$$X_L = 5 \Omega$$

$$X_C = 2 \Omega$$

设 b 节点的电位为 0, a 节点的电位为 U_a

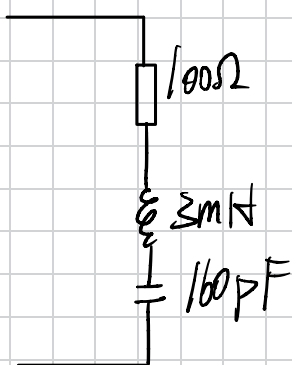
对于 a 由 KCL 定律

$$\frac{U_a - U_{S1}}{-jX_C} + \frac{U_a - U_{S2}}{jX_L} = \frac{U_a}{R}$$

$$\text{得 } U_a = -100 + 100j$$

$$\dot{I}_R = \frac{U_a}{R} = -20 + 20j = 20\sqrt{2} \angle -45^\circ \text{ A}$$

3-13



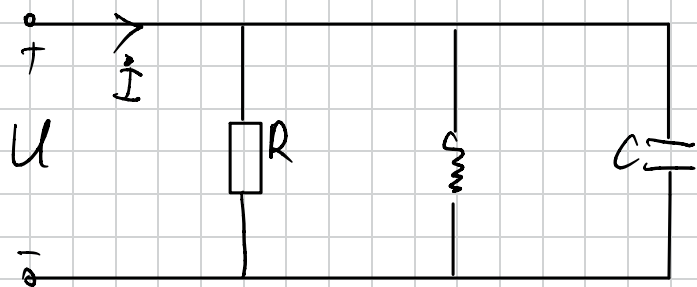
$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{3 \times 10^{-3} \times 160 \times 10^{-12}}} = \frac{1}{8\sqrt{3}\pi \times 10^{-7}} \text{ Hz}$$

$$Q = \frac{U_C}{U} = \frac{U_0 L}{R} = \frac{2\pi f_0}{100} = 43.3$$

$$I = \frac{U}{R} = \frac{1.5}{100} = 0.015 \text{ A}$$

$$U_L = U_C = 1.5 \times 43.3 = 64.95 \text{ V}$$

3-15



$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{40 \times 10^{-6} \times 40 \times 10^{-12}}} = \frac{1}{80\pi \times 10^{-9}}$$



$$Q = \frac{R}{U_0 L}$$

$$\Rightarrow R = Q U_0 L = 60 \times 25 \times 10^6 \times 40 \times 10^{-6} = 60 \times 10^3 \Omega$$

$$U = R I_0 = 60 \times 10^3 \times 0.3 \times 10^{-3} = 18 \text{ V}$$